THseries
Industrial Robot

TRANSPORTATION and INSTALLATION MANUAL

Notice

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

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Shibaura Machine

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Preface

This manual describes how to unpack and install the TH series industrial robot and controller. Specifically, it describes how to unpack a shipment containing the equipment, how to connect wiring and air piping, and how to attach tools. Be sure to look through this manual before unpacking the shipment.

Before beginning the work according to this manual, read through the Introductory Manual and Safety Manual so that you can learn the names and functions of various components and the safety measures.

This manual is divided into the following four (4) sections:

Section 1 Transportation
This section describes how to remove the robot and controller from their boxes and how to transport them to the installation site. This section also discusses how to temporarily store the equipment after unpacking the shipment.

Section 2 Installation
This section discusses the equipment installation environment, space requirements, and how to install the equipment.

Section 3 System Connections
This section describes how to connect the robot, controller and peripheral equipment.

Section 4 Tool Interlace
This section describes how to connect the tool to the robot arm and how to connect pipes and wires to the tool. This section also discusses maximum permissible loads of the tool.

Section 5 Specifications

Precautions on Safety

Important information on the robot and controller is noted in the instruction manual to prevent injury to the user and persons nearby, prevent damage to assets and to ensure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 Danger	This means that "incorrect handling will lead to fatalities or major injuries".
 Caution	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)".

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long term treatment.

*2) Physical damage refers to major fires due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited(must not be done). The details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory(must be done). The details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger and caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

Caution

- Always read through the Safety Manual provided separately before starting actual work to ensure safety work covering from the robot installation to operation.

[Installation and transportation]

Always observe the following items to safely use the robot.

Danger	
Prohibited 	• DO NOT install or operate if any parts are damaged or missing. Doing so could lead to electric shocks, fires or faults. • DO NOT install the robot where it may be subject to fluids such as water. Doing so could lead to electric shocks, fires or faults. • Do not place the robot near combustible matters. Doing so could lead to fires if the matter ignites due to a fault, etc.
 Mandatory	• Always secure the robot with the attached clamps before transporting it. Failure to do so could lead to injuries if the arm moves when the robot is suspended. • Wire the robot after installation. Wiring the robot before installation could lead to electric shocks or injuries. • Always use the power voltage and power capacity designated by SHIBAURA MACHINE. Failure to do so could lead to device faults or fires. • Always use the designated power cable. Using a cable other than that designated could lead to fires or faults.
 Always ground	• Completely connect the grounding cable. Failure to do so could lead to electric shocks or fires if a fault or fault current occurs. Noise could lead to malfunction. Also, it could cause mis-operation by noise.

Caution	
 Prohibited	• NEVER lift the robot by the arm 2 cover or arm 2. Doing so will apply an excessive force on the robot's mechanism section and could lead to faults. • For the controller, secure the ample space for air vent. Heating of controller could lead to malfunction.
 Mandatory	• When lifting the robot, lift it up slowly. The robot will tilt slightly, so lifting it up suddenly could be hazardous. • When storing the robot, securely fix it on the base. The robot will be unstable if just set down, and it could tilt over.

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Section 1

Transportation

1.1 Unpacking

The robot and controller are shipped separately in wooden crates and corrugated cardboards. The teach pendant and other accessories such as the instruction manual are packaged separately.

Open the packages in a location easily accessible to where the equipment is to be installed. Take care not to damage the robot or controller. After opening the packages, check to make sure that all the accessories are present and that nothing has been damaged in transit.

Danger

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden crate, corrugated cardboard, polyethylene shipping bag and cushion material according to your standards.

1.2 Transportation

Move the robot and controller very carefully. Make sure that no excessive impact ($9.8 m/s^2$ or more) or vibration is applied on the equipment. If the equipment is to be subject to vibration over a long period, be sure to tighten all the nuts and screws.
If the equipment is to be moved to a location some distance from where it was unpacked, put the equipment back into the wooden crates and corrugated cardboards and reposition the cushions as they were.

1.2.1 Mass and Dimensions

The mass and dimensions of the robot and controller are shown in Table 1.1.

Table 1.1

Model	Mass
Robot TH250	15 kg
Robot TH350	16 kg
Controller TS1000	10 kg

1.2.2 Transporting the Robot

In principle, transport the robot in the posture as shown in Fig. 1.1. The robot is shipped with the arm folded back and secured with a clamp. (After you have unpacked the shipment, you can move it as it is.)

⚠ Danger

- Be sure to secure the arm with the attached clamp before transporting the robot. Failure to do so could cause a hazardous situation as the arm will move when the robot is suspended.

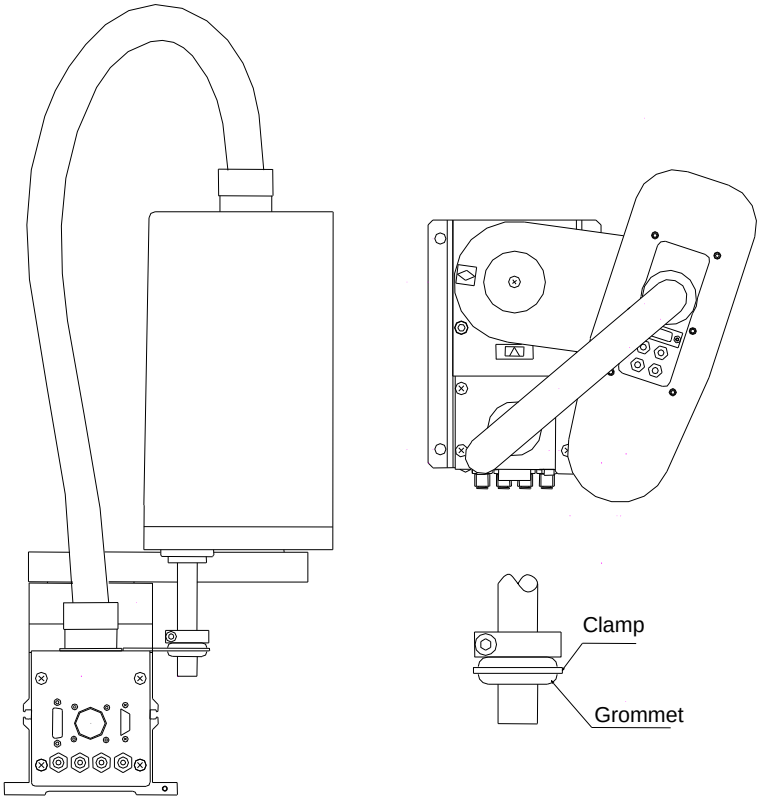


Fig. 1.1 Transportation posture

Caution

- Both the robot and robot controller weigh heavy. When transporting and unpacking them, take careful precautions not to drop them. Otherwise, you may be seriously injured or the equipment is damaged heavily.
- When lifting the robot by workers, hold the positions shown in Fig. 1.2 by hands. If the arm 2 or cover is held by hands, an unusually large force is exerted, resulting in a trouble.
- If the robot has been carried manually by the workers, take utmost care to prevent his hand or leg from being caught in the robot when placing it on the floor.

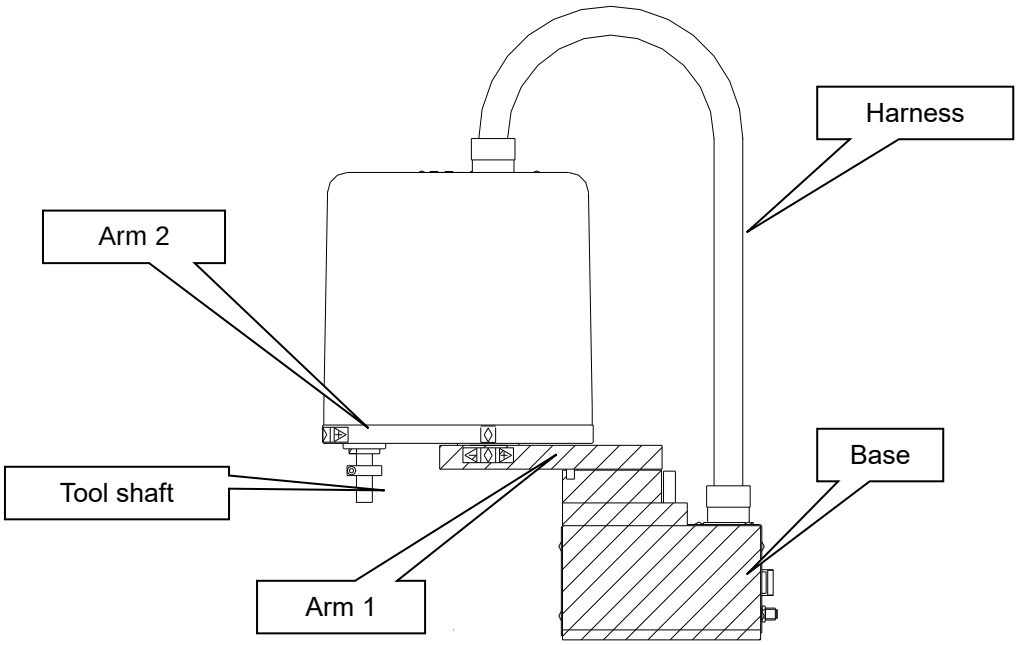


Fig. 1.2 Robot handling places (shaded area)

After installing the robot, remove the clamps.

1.2.3 Transporting the Controller

Disconnect all cables and teach pendant before transporting the controller.

Danger

- When placing the controller on the floor, etc., make sure not to catch your hands or feet.

1.3 Storage

We do not recommend storing the robot and controller for long periods of time after unpacking them. If this is unavoidable, however, strictly observe the following precautions for storage.

1.3.1 Storage Precautions for the Robot

⚠ Caution

- Secure the base completely to prevent the robot from falling over. When placed as it is, the robot is not stable and will fall over.
- Keep the robot out of direct sunlight. The timing belts and resin covers may deteriorate.
- Seal the robot in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture. As the tool shaft is susceptible to rust development, coat the tool shaft with grease and the other surfaces with a rust preventative compound.

1.3.2 Storage Precautions for the Controller

⚠ Caution

- Keep the controller out of direct sunlight. Otherwise, the controller interior will be excessively heated up, causing a trouble.

Section 2

Installation

2.1 Installation Environment

Table 2.1 shows the environmental conditions for the location in which the TH series Robot and TS1000 controller are to be installed.

Table 2.1 Environmental Conditions for the Robot and Controller

Item	Specifications
Temperature	In operation : 0 to 40 °C In storage : –10 to 50 °C
Humidity	20 to 90 % (Non-condensing) DO NOT install the robot where it may be subject to fluids such as water.
Altitude	1000 m or less
Vibration	In operation : 0.98m/s^2 or less In transport : 9.8m/ s^2 or less
Dust	No inductive dust should exists. Consult with SHIBAURA MACHINE first should you wish to use the robot and controller in an especially dusty environment.
Gas	No corrosive or combustible gas should exist.
Sunlight	The robot must not be subject to direct sunlight.
Electromagnetic noise	A heavy noise source should not exist nearby.
Magnetic field	A heavy magnetic field source should not exist nearby.

Danger

- Do not place the robot or controller near combustible. Doing so could lead to fires if it ignites due to a fault, etc.

⚠ CAUTION

- In the case where batteries for detecting the motor position are of alkaline type, the batteries can overheat, leak battery fluid, or rupture when used under high temperatures. Also, high temperatures can reduce the performance and lifespan of the battery. If using the robot under high temperatures, please consult with the SHIBAURA MACHINE sales office.

2.2 Robot Installation

Before installing the robot, you should plan a layout, fully considering the working envelope, coordinate system and space for maintenance.

2.2.1 External Dimensions

External view drawings of the robot are shown in Figs. 2.1 and 2.2.

2.2.2 Working Envelope

Each axis can operate within the working envelope. To keep the robot from moving out of the working envelope by mis-operation, the robot is equipped with mechanical stoppers outside the working envelope. Additionally, soft limits are provided which may be set as desired by the user.

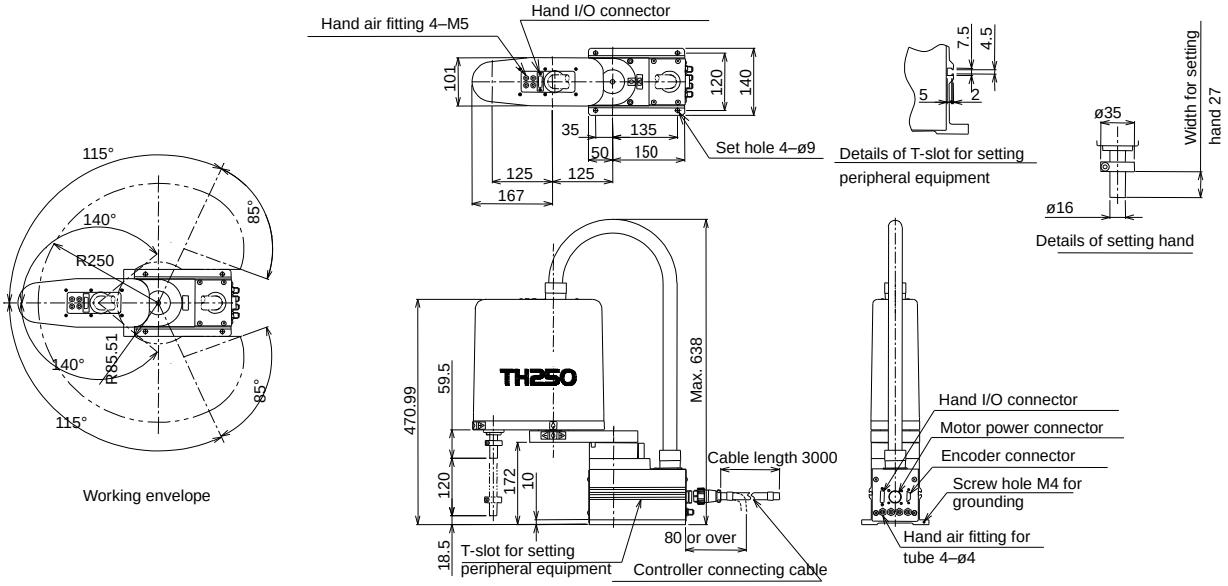


Fig. 2.1 TH250 external view

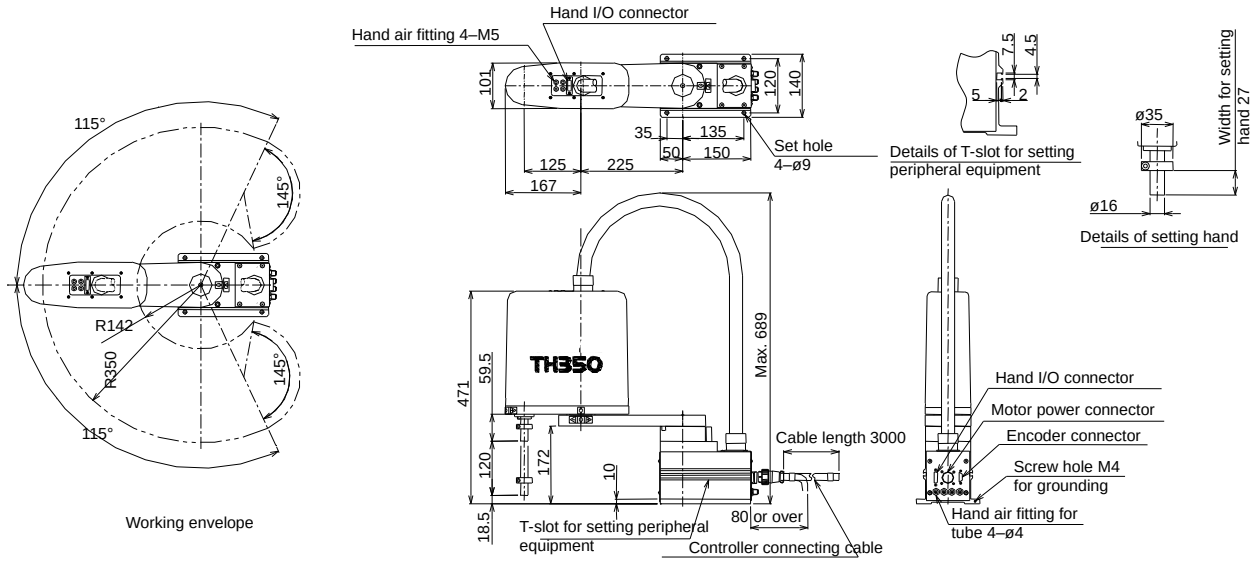


Fig. 2.2 TH350 external view

2.2.3 Coordinate System

The robot's joint angle zero point (0° or 0mm position) is factory-calibrated according to the base reference planes. The base coordinate system is determined according to this calibration. Figs. 2.3 and 2.4 show the base coordinate system and zero position of each axis joint angle.

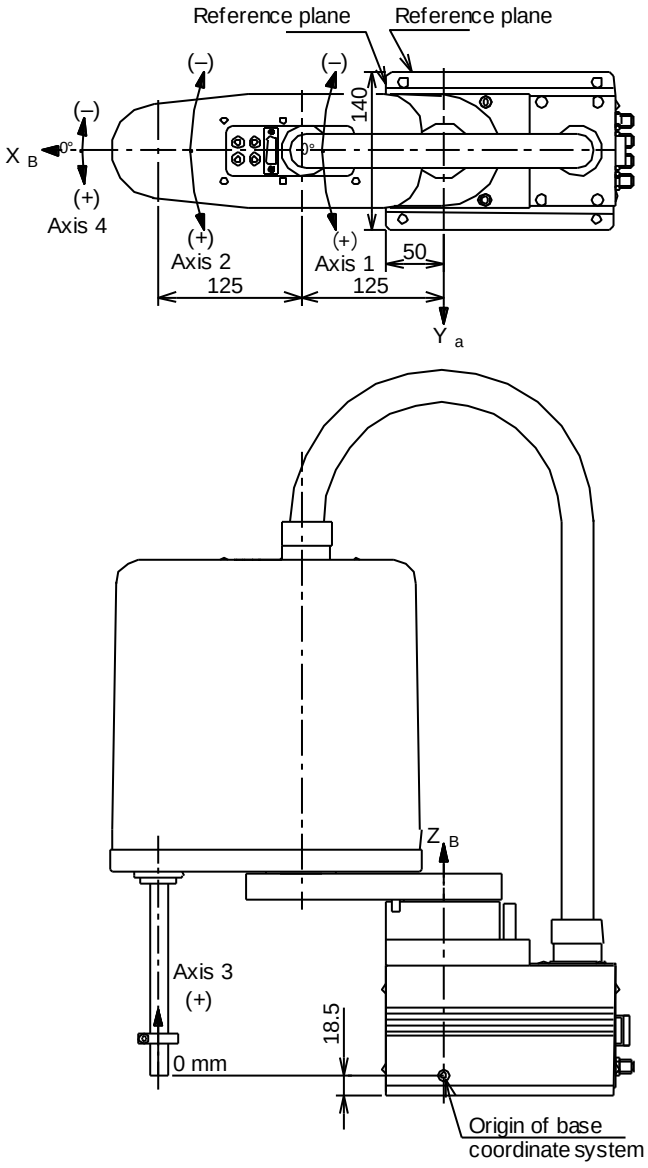


Fig. 2.3 TH250 base coordinate system and joint angle zero point

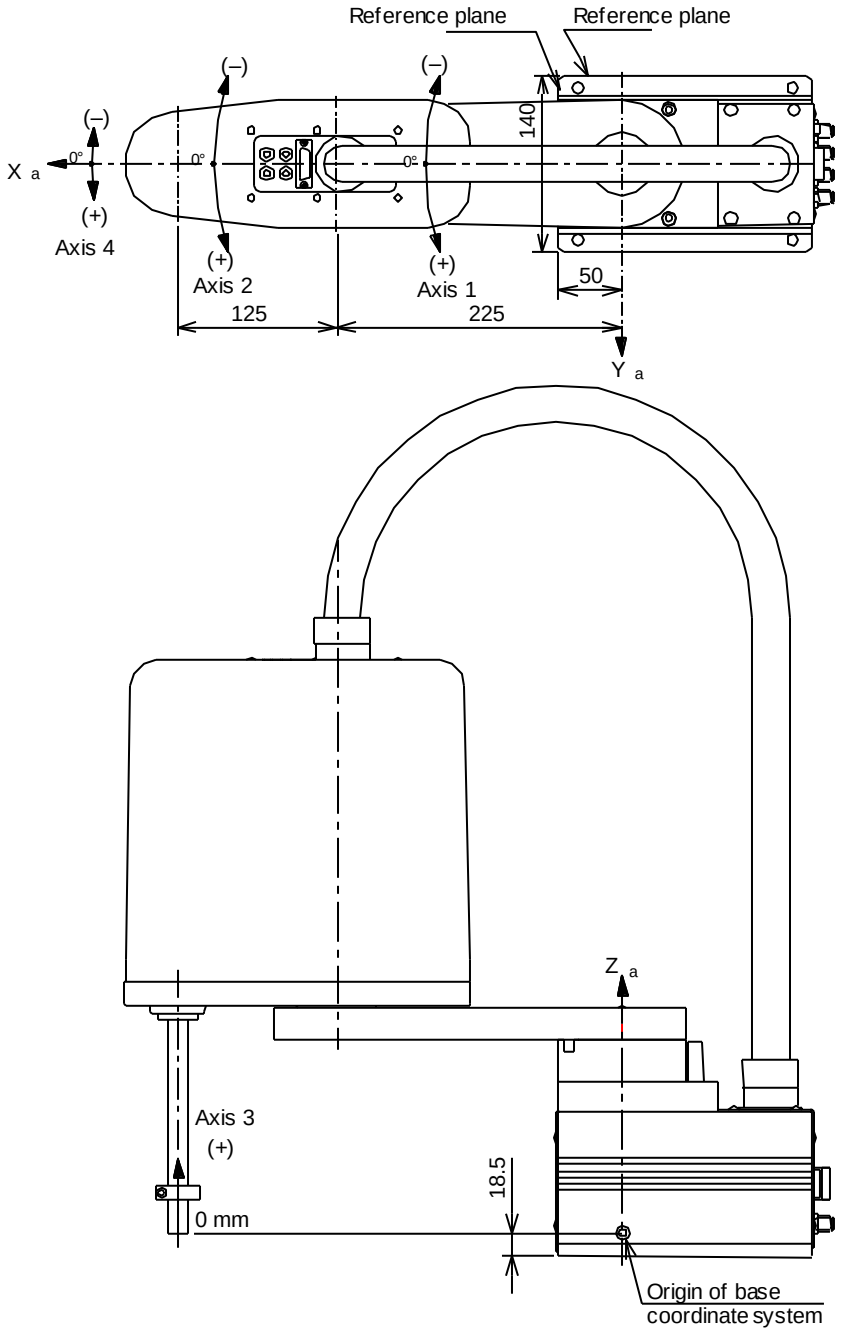


Fig. 2.4 TH350 base coordinate system and joint angle zero point

2.2.4 Installing the Robot

The robot is secured, using the set holes on the base. Use M8 hexagon socket head cap screws.

The robot installation method is shown in Fig. 2.5. Reference planes are prepared on the base section and are marked with "xxx".

To align the robot's position in the base coordinate system, or to replace the robot, prepare adequate reference planes. Then, contact such reference planes to the base reference planes and secure the robot.

⚠ ıtion

- The robot will suddenly accelerate and decelerate during operation. When installing it on a frame, make sure that the frame has sufficient strength and rigidity.

  If the robot is installed on a frame that does not have sufficient rigidity, the vibration will occur while the robot is operating, and could lead to faults.

  When installing the robot on the floor, secure the robot with anchor bolts, etc.
- Install the robot on a level place. Failure to do so could lead to a drop in performance or faults.

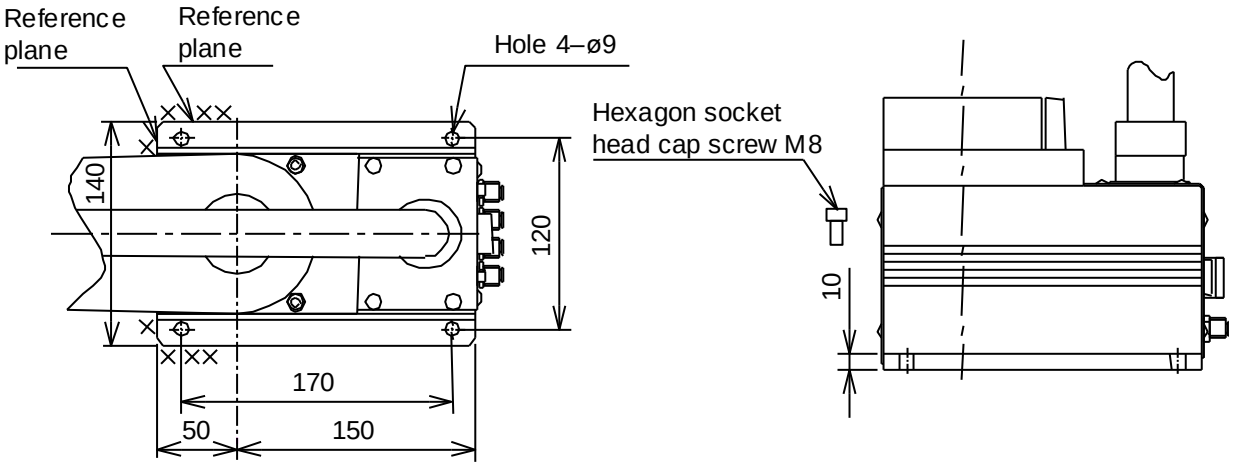
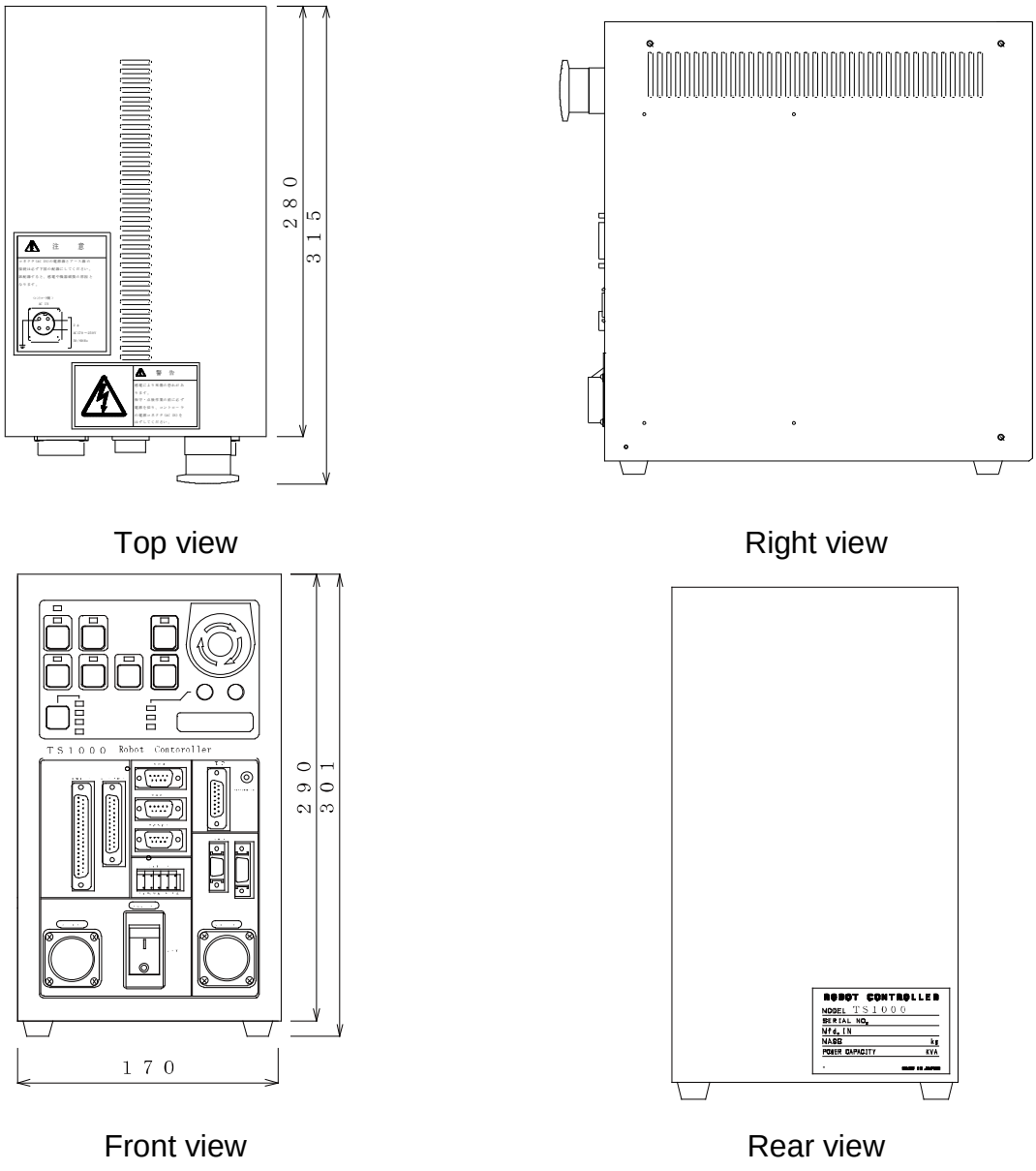


Fig. 2.5 Installing the TH250 and TH350 robots

2.3 Installing the Controller

2.3.1 External Dimensions

An external view of the controller is shown in Fig 2.5.



Top view Right view

Front view Rear view

Fig. 2.5 TS1000 controller external view

2.3.2 Rack Mounting Dimensions

When mounting the TS1000 robot controller in a rack, set the side brackets using the screw holes provided on both ends of the front panel, and secure the controller to the rack. The side brackets [1] in Fig. 2.6 are optional.

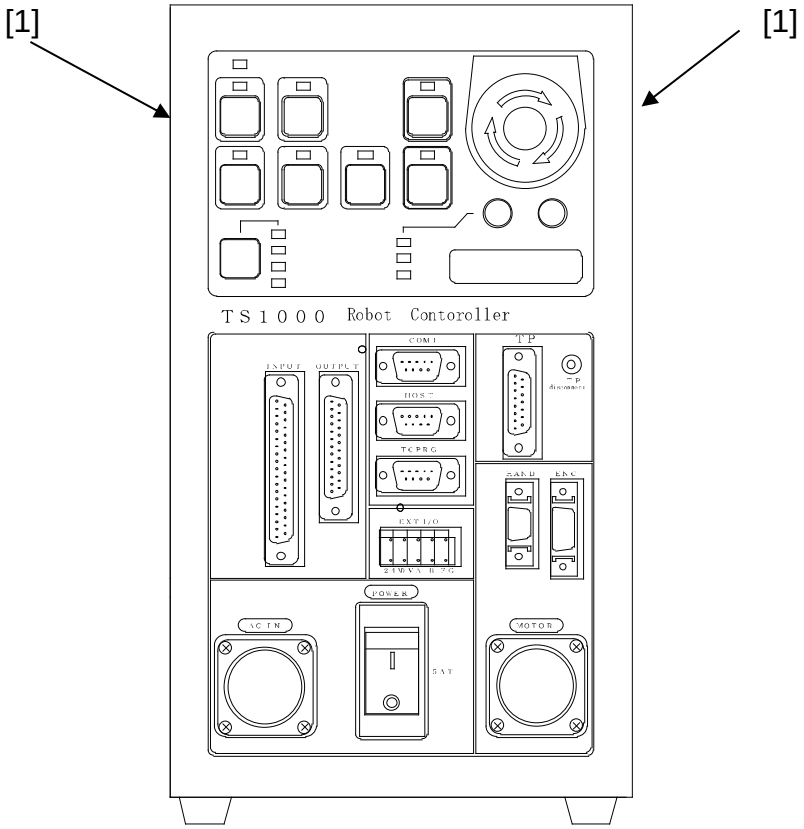


Fig. 2.6 Dimensions of controller rack mounting holes

2.3.3 Precautions for Rack Mounting

Pay attention to the following matters when rack-mounting the TS1000 robot controller.

(a) When mounting the controller in a rack, use the screw holes provided at both ends of the front panel, and secure the controller. (Optional side brackets are required.)

⚠ Caution

- If the rack is completely sealed, provide holes to allow the air to be let out, force-ventilate the rack with a fan, or cool it indirectly so that the heat will not be trapped in the rack.

  If the heat is trapped in the rack or controller, faults could occur.

(b) As the cable connectors are connected to the front side of the controller, provide a space of 130 mm on the front side.
For maintenance, the upper cover should be removed. (See Fig. 2.7.)

Keep this in mind when installing the controller. Specifically, the controller should be easily removable from the rack. Practically, be careful of the following points.

1) Arrange the cables around the front panel of the controller (so that the controller can be removed).
2) Arrange the cables between the controller and control panel when the control panel is separated.
3) Connect all cables in such a manner that the robot can be operated even if the controller is removed from the rack.

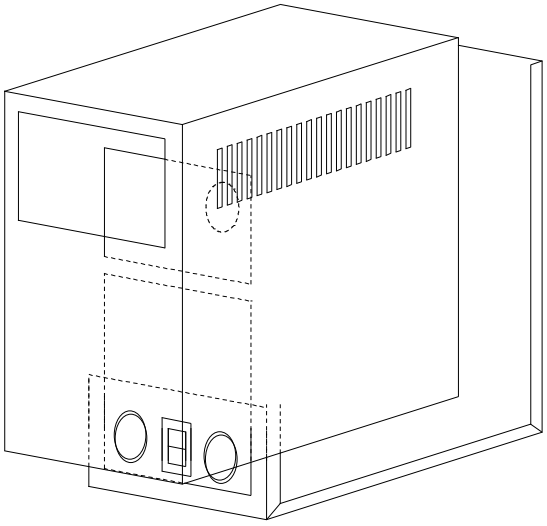


Fig. 2.7 Removing upper cover

(c) When the controller is mounted on the rack, the weight of the controller should be supported with the legs of the controller. Screw holes for rack-mounting the controller are used for securing the controller panel, and the weight of the controller cannot be supported only with the screws.

(d) On the front of the controller, a clearance of approx. 130 mm should be provided for connecting the connector of the teach pendant. Even if the teach pendant is not used, a clearance of approx. 50 mm is required for connecting a dummy plug.

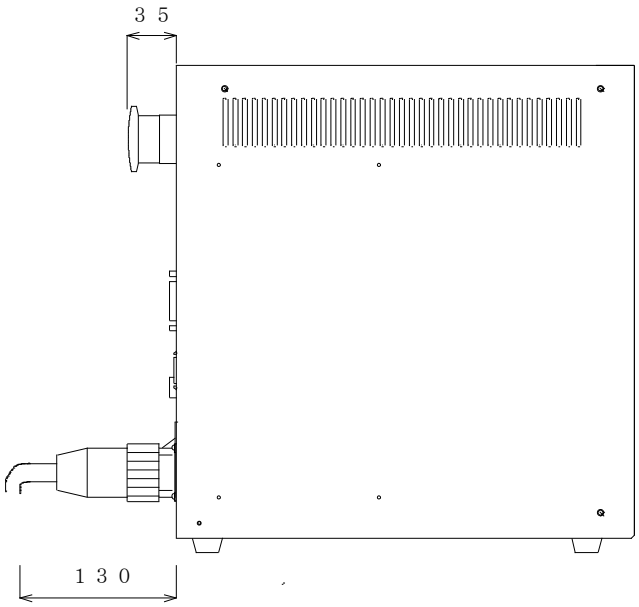


Fig. 2.8 Clearance of controller front panel

2.3.4 Precautions for Direct Installation

When two (2) or more controllers are used, it is necessary to provide a clearance of 50 mm or more in the horizontal direction and a clearance of 100 mm or more in the vertical direction between the controllers.

⚠ution

- Provide a ventilation space at the side of the controller so that the air vent holes are not blocked. The space equal to the length of the legs should be kept below the lower surface.

  Failure to do so could cause the cooling performance to drop or to faults.
- DO NOT stack the controllers. DO NOT place any object on top of them.

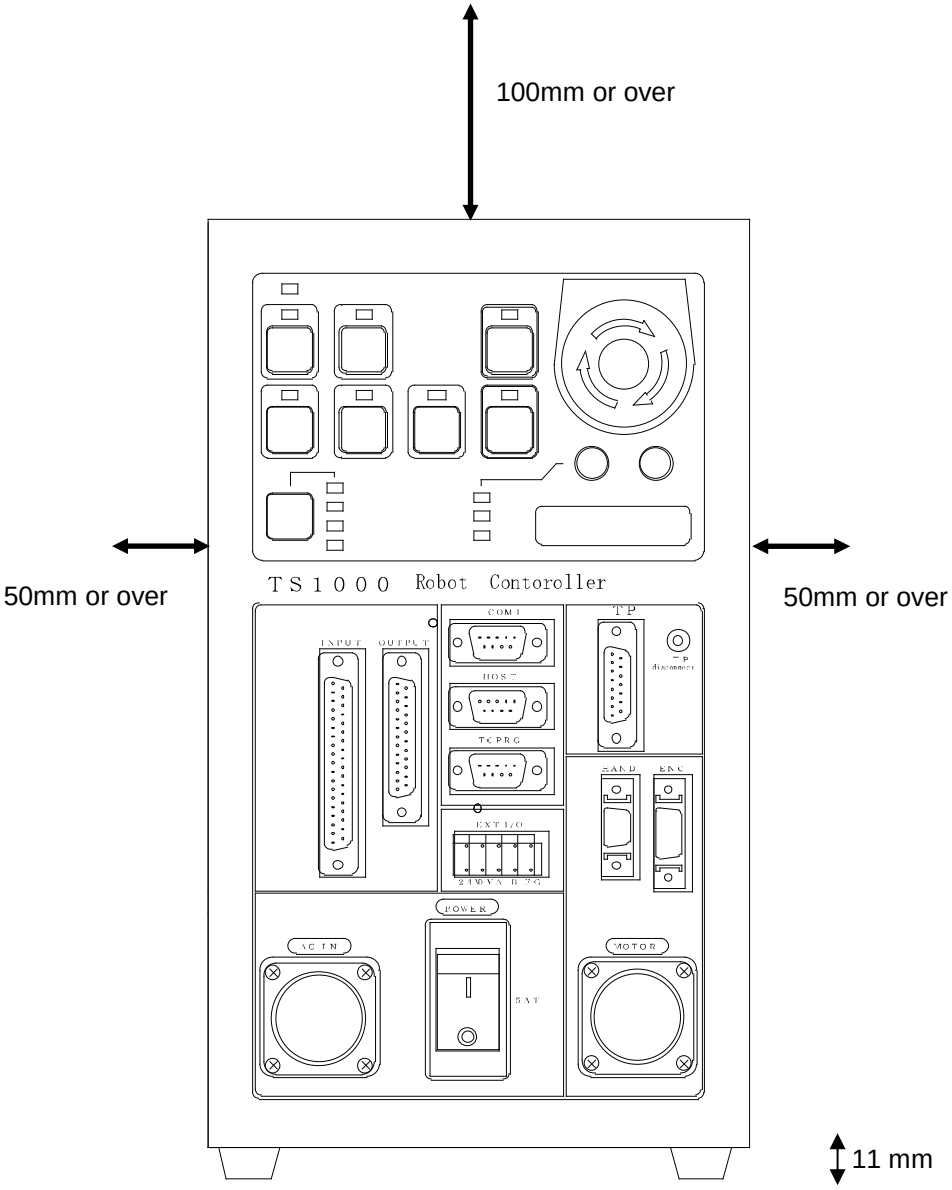


Fig. 2.9 Controller ventilation space

2.4 Precautions for Handling the Teach Pendant

Be careful of the following matters when handling the teach pendant.

⚠ution

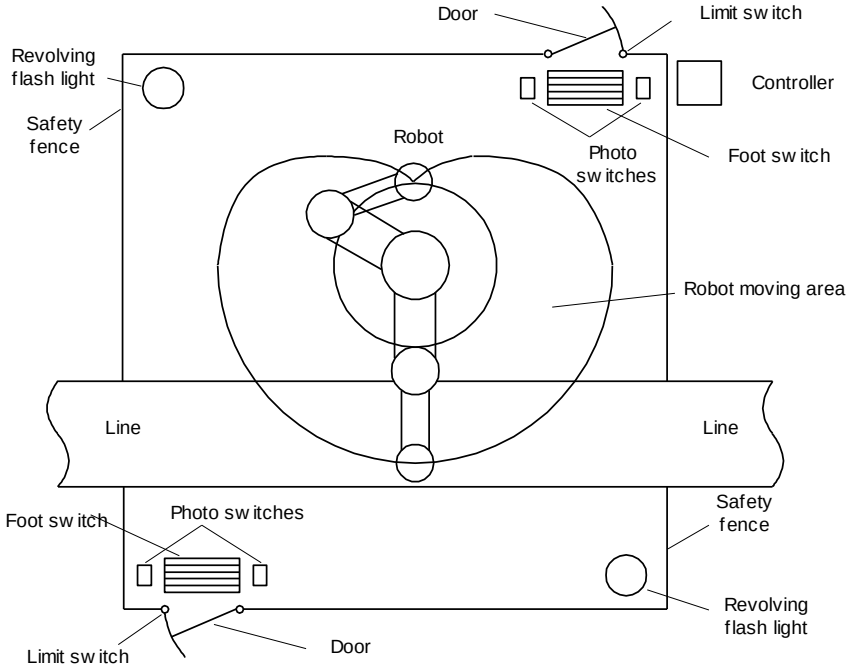
- DO NOT drop the teach pendant or hit it against anything.
- DO NOT pull the cable running from the teach pendant.
- DO NOT press the switches on the teach pendant with anything sharp (like the tip of a knife, pencil, ball-point pen, etc.).
- DO NOT place or use the teach pendant near open flames.
- DO NOT leave the teach pendant in direct sunlight for a long period of time.

2.5 Safety Measures

a) When installing the robot, provide sufficient space to carry out the work safely.

b) Clarify the hazard zone, and provide a safety fence so that other persons cannot enter the zone easily. The hazard zone is the zone near the robot's working space where a hazardous state could occur if a person enters.

c) Provide a limit switch, photo switch or foot switch, etc., at the entrance of the safety fence to provide an emergency stop function that will stop the robot if a person enters the hazard zone. The emergency stop function should be an electrically independent close contact (closed in normal operation) with compulsive opening function and must not be automatically recovered.



d) The controller should be installed at a place outside the hazard zone where the operator can view the robot movement.

2.6 Position Detector Backup Batteries

The robot has batteries to back up positional information on position detectors in its base. Unless the robot is used for a long period of time, the battery voltage will drop and the positional data of zero point will be lost.
Also, if the battery voltage for backup of the X8LC printed board SRAM of controller has dropped, absolute position data set by the user will be lost. To avoid this, replace each battery on a regular basis.

If the battery line voltage has dropped, an alarm of level 1 as shown below occurs. When this happens, replace the battery immediately.

Error code	Descriptions
1–049	Axis1 Battery alarm
1–065	Axis2 Battery alarm
1–081	Axis3 Battery alarm
1–097	Axis4 Battery alarm
1–113	Axis5 Battery alarm
1–129	Axis6 Battery alarm
1–145	MAIN Battery alarm

If the line voltage has dropped, an alarm of level 8 as shown below occurs. When this happens, replace the battery with a new one and adjust the zero point of the robot, if shifted, because there is a fear that the position encoder backup data has been lost.

Error code	Descriptions
8–065	Axis1 Encoder abnormal
8–097	Axis2 Encoder abnormal
8–129	Axis3 Encoder abnormal
8–161	Axis4 Encoder abnormal
8–193	Axis5 Encoder abnormal
8–225	Axis6 Encoder abnormal

Section 3

System Connections

3.1 Cable Wiring

This section describes the various types of cables and connectors and explains how these are to be connected.

3.1.1 Connector Arrangement on the Controller

The cables connected to the TS1000 robot controller are shown in Fig. 3.1.

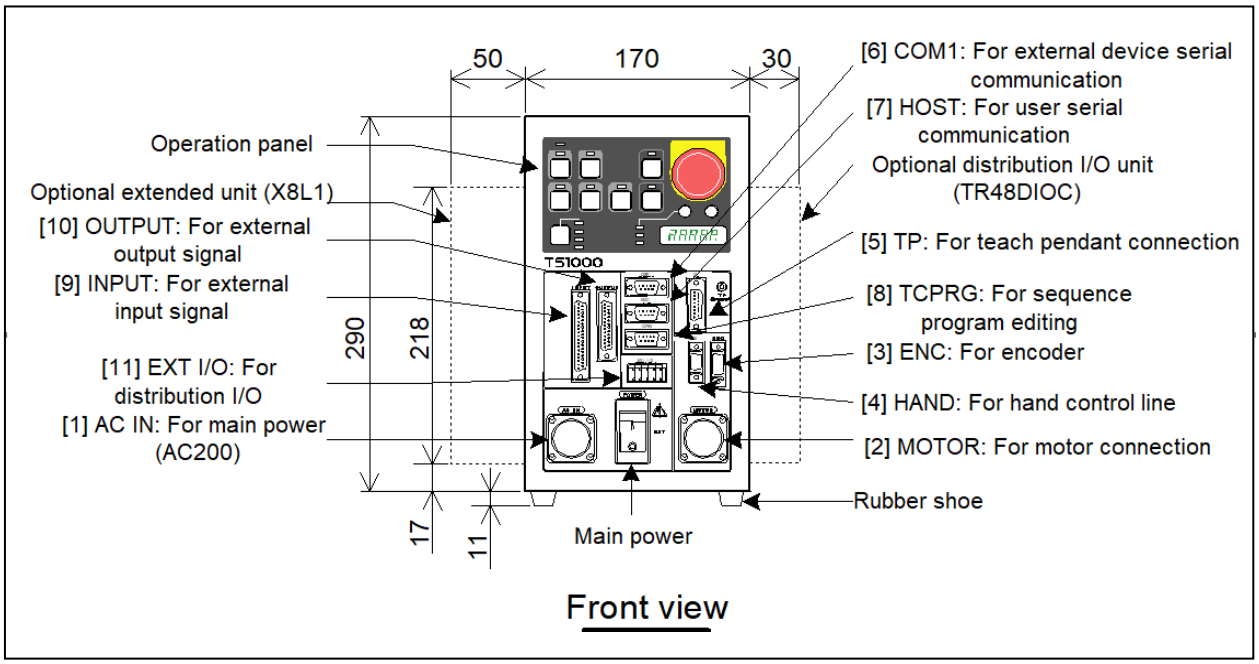


Fig. 3.1 Robot controller connector layout

In this Section, we explain how to connect cables [1] to [5]. For information on how to connect cables [6] and [7], refer to the Interface Manual.

3.1.2 Connecting the Power Cable "ACIN"

([1] of Fig. 3.1; plug connector attached)

The power cable is used to supply the main AC power to the controller.

Table 3.1 Power supply specifications

Item	Specifications
Power supply	Single phase, AC 200/220 V ±10 %, 50/60 Hz± 1 Hz
Power supply capacity	TH250 (1.5 kVA) TH350 (1.7 kVA)
Instantaneous power failure	1/2 cycle or less
Grounding	Grounding with grounding resistance of 100 Ω or less

The connector is ACIN ([1] of Fig. 3.1)
The cable is not an accessory, so use the attached plug connector connected to ACIN on the controller side to manufacture a cable.

ACIN plug connector	Type: JL04V–6A18–10SE–EB	Maker:	Japan Aviation Electronics Industry
ACIN cable clamp	Type: JL04–18CK	Maker:	Japan Aviation Electronics Industry
Wire	3.5 mm^2 ~ 5.5 mm^2		

Wires are to be soldered to the connector (JL04V–6A18–10SE–EB).

⚠ Danger

- Always use the designated wire. Failure to do so could lead to fires or faults.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
- After making the connection, use a tester, etc., to confirm the connection.

For the terminal arrangement, see Para. 3.1.7.

⚠ution

- Unless the main power is normally supplied to the controller due to phase defect or voltage drop, an error of 8–027 “Slow Charge error” occurs. When this happens, make sure that the line voltage at the controller power connector satisfies the specified input power of the controller, and that the same line voltage is stabilized.
- For details of the 8–027 error, see Para. 13.8 of the Operator’s Manual.

3.1.3 Connecting the Motor Cable "MOTOR"

([2] of Fig. 3.1) (Cable attached)

The motor cable connects the controller and robot and supplies the power required to rotate the motor from the controller's servo driver to each robot motor. It also turns on and off the motor brake. The motor cable runs from [2] of Fig. 3.1 (MOTOR) to [1] of Fig. 3.2.

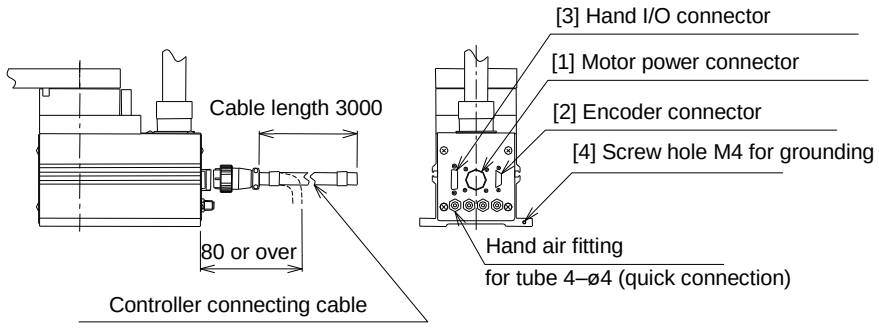


Fig. 3.2 Robot side connector arrangement

3.1.4 Connecting the Encoder Cable "ENC"

([3] of Fig. 3.1) (Cable included)

The encoder cable is a signal line used to transmit data from the encoders (devices used to detect the revolution (and thereby the position) of each shaft) to the controller. The encoder cable runs from [3] of Fig. 3.1 (ENC) to [2] of Fig. 3.2.

3.1.5 Connecting the Hand Control Signal Cable "HAND"

([4] of Fig. 3.1) (Cable: Option)

The hand control signal cable is used for input and output of robot control signals such as hand operation. The hand control signal cable runs from [4] of Fig. 3.1 (HAND) to [3] of Fig. 3.2.

3.1.6 Grounding the Robot

An M4 screw hole for grounding the robot is provided on the robot base. (Fig. 3.2 [4]) Be sure to connect this cable with the conductor for the entire facility.

3.1.7 Inserting and Removing Cables

⚠ Caution

- Before inserting or removing any controller cable, be sure to turn off the "POWER" switch.
- When removing a cable, be sure to pull the plug and not the cord. Otherwise, you may damage the cable.

a) Circular connectors: ACIN, MOTOR

First, completely insert the connector on the cable side into the controller connector. Then, turn the cable side lock screw to the right to clamp the cable. A loose plug can cause an accident or failure, so be careful and do the job right. To remove a connector, turn the lock screw to the left and pull out the cable side connector.

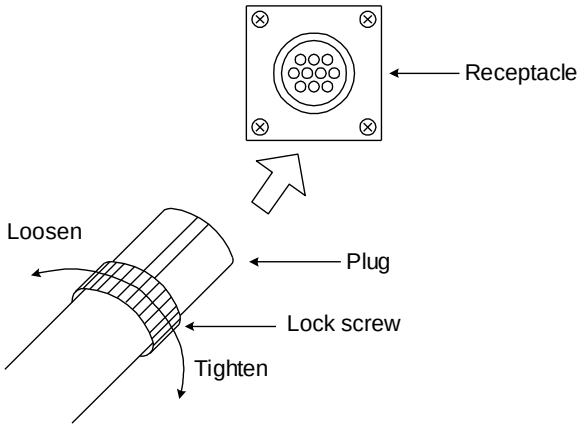


Fig. 3.3 Inserting and removing a circular connector

b) Square connectors: ENC, HAND, TP, INPUT, OUTPUT, COM1, HOST1, TCPRG

First, completely insert the cable side connector into the controller connector. Then, tighten the lock screws on both ends of the cable side connector with a screwdriver. A loose plug can cause a contact failure or other accident, so be careful and do the job right.

To remove the connectors, first loosen the lock screws, then pull out the cable side connector.

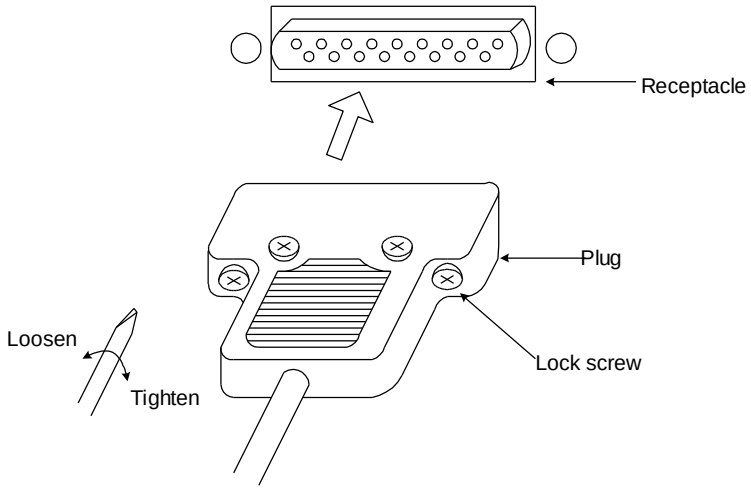
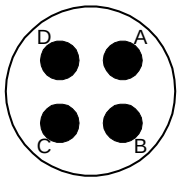


Fig. 3.4 Inserting and removing a square connector

3.1.8 Examples of Connector Terminal Arrangement

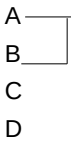
a) Power cable connector ACIN



Connects to controller.

Type: JL04V–2E18–10PE–B

Manufacturer:
Japan Aviation
Electronics Industry



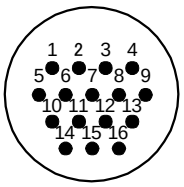
Single phase, AC 200/220 V ±10 %, 50/60 Hz

Danger

- Completely connect the grounding cable.
  Failure to do so could lead to electric shocks or fires if a fault or fault current occurs.

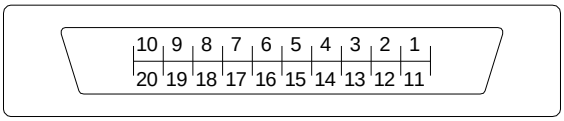
Grounding with grounding resistance of 100 Ω or less.

b) Motor cable connector MOTOR



Connects to controller.
Type: JR25RK-16S
Manufacturer: Hirose Electric

c) Encoder cable connector ENC

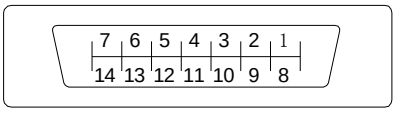


Connects to controller.

Type: 54306–2011

Manufacturer: Molex Co., Ltd.

d) Robot control signal cable connector HAND

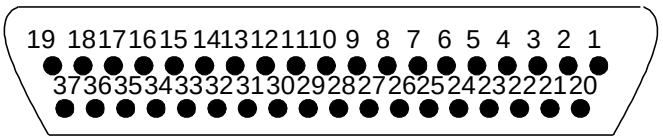


Connects to controller.

Type: 54306–1411

Manufacturer: Molex Co., Ltd.

e) External input signal cable connector INPUT

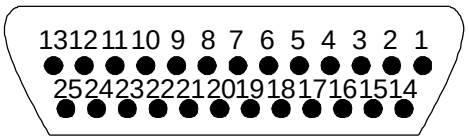


Connects to controller.

Type: XM2F-3740-112L

Manufacturer: OMRON Co.,Ltd

f) External output signal cable connector OUTPUT



Connects to controller.

Type: XM2F-2540-112L

Manufacturer: OMRON Co.,Ltd

3.2 Controller Connector Signals

3.2.1 Connector Signal Connection Diagrams

Diagrams showing which signals correspond to which terminals are shown in Section 2 of the Interface Manual.

3.2.2 Jumpers for Safety Related Signals

The following external control input signals are provided to serve for the safety purpose.

External control input signals
INPUT–12 (STOP)
INPUT–13 (LOW_SPD)
INPUT–14 (SV_OFF)
INPUT–31 (CYCLE)
INPUT–32 (BREAK)
INPUT–18, 19 (EMS1B ~ EMS1C)
INPUT–36, 37 (EMS2B ~ EMS2C)
INPUT–17, 35 (INCOM3 ~ P24G)

These signals are already jumpered for the connectors provided with the controller. If you wish to use these signals, therefore, you should remove the jumpers and rewire as appropriate. Also, if you plan to use the robot without external signals, be sure to connect the attached connector to the controller side INPUT connector.

Connector Jumpers

INPUT			
12–16	14–16	32–16	36–37
13–16	31–16	18–19	

Caution

Unless both the SVOFF and emergency stop contact 1/2 signals are jumpered, the controller servo power cannot be turned on.

Section 4

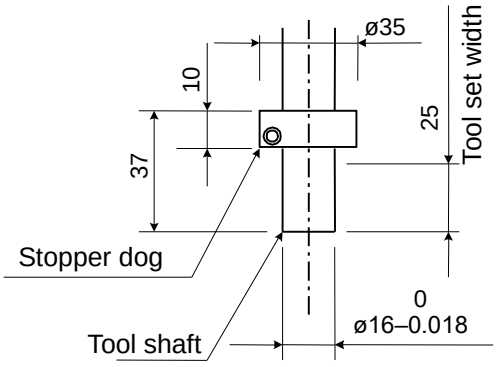
Tool Interface

4.1 Mounting the Tool (Hand)

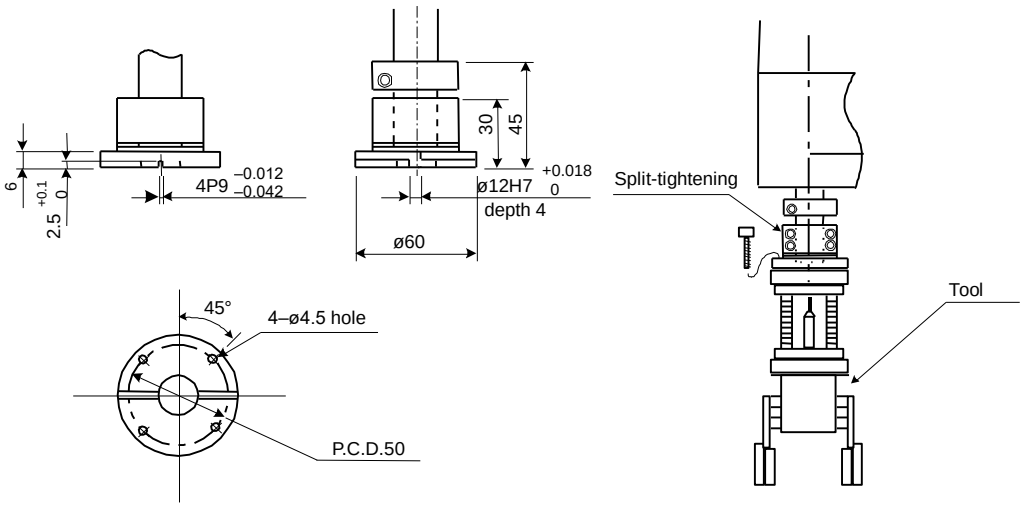
(1) TH250, TH350

The tool is mounted on the end of the tool shaft (ø16). The tool shaft has a dog for the stopper, and the section in front of the dog is used for mounting the tool. At this time, the dog should not be removed. Dimensions of the tool shaft section are shown in Fig. 4.1 (a).

As shown in Fig. 4.1, the tool is secured to the tool shaft with split clamp, but since there is no reference such as keyway on the tool shaft, take care of the tool direction when mounting.



a) Tool shaft section dimensions



b) Dimensions of tool set flange (option)

Fig. 4.1 Mounting the TH250 and TH350 tools

4.2 Tool (Hand) Wiring and Piping

The robot is provided with wiring and air piping for the tool. These wiring and piping extend from base section to the arm 2.

a) Tool wiring

Arm side
DSUB 25-pin connector
（XM2D–1501: OMRON）

Base side
DSUB 25-pin connector
（XM2A–1501: OMRON）

Pin No.	Signal name	Pin No.	Signal name
1	LINE1	1	LINE1
2	LINE2	2	LINE2
3	LINE3	3	LINE3
4	LINE4	4	LINE4
5	LINE5	5	LINE5
6	LINE6	6	LINE6
7	LINE7	7	LINE7
8	LINE8	8	LINE8
9	LINE9	9	LINE9
10	LINE10	10	LINE10
11	LINE11	11	LINE11
12	LINE12	12	LINE12
13	LINE13	13	LINE13
14	LINE14	14	LINE14
15	Shield	15	Shield

Fig. 4.2 Machine I/O harness wiring

b) Tool air piping

A total of four (4) air pipelines (ø4 × 2.5) are provided for the tool, which are shown in Fig. 4.3 below.

An air control unit (i.e., oiler, regulator with gage, filter) of the solenoid valve should be provided by the user.

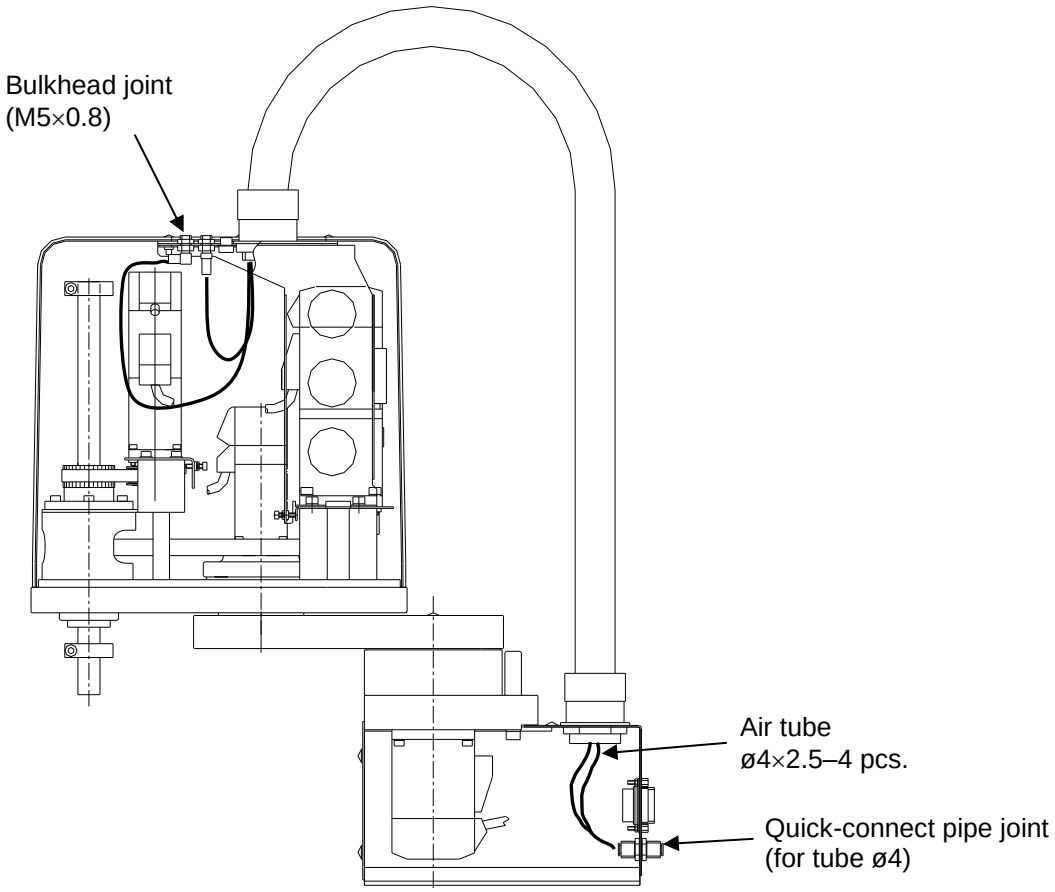
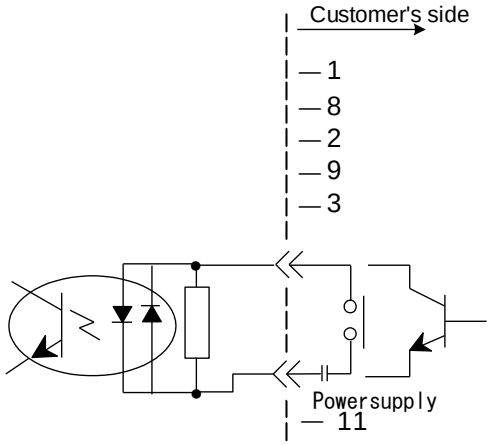


Fig. 4.3 Tool air piping

4.2.1 Controller's Tool Signals (TS1000)

The controller is provided with the tool signals (i.e., five (5) input signals for sensor, etc., four (4) control signals for solenoid valve, etc., DC 24V and DC 24 VGND signals), which can be connected also from the controller side. (A cable between the controller and cable is an option.) The relevant signals are described below.

a) Input signal connector HAND

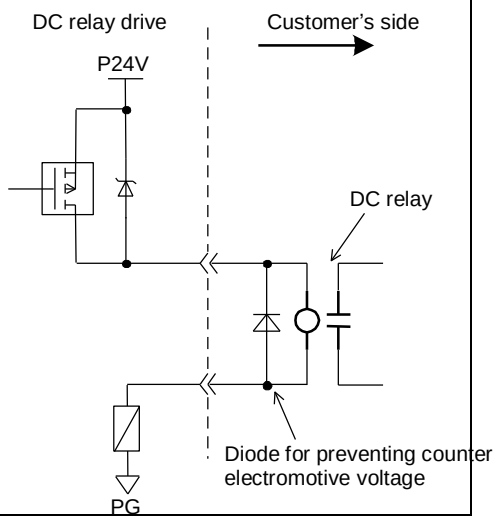
Pin	Signal name		Signal No.	Input circuit and example of connections
1	HI_1	Hand input 1	201	
8	HI_2	Hand input 2	202	
2	HI_3	Hand input 3	203	
9	HI_4	Hand input 4	204	
3	HI_5	Hand input 5	205	
10	P24V	24 V		
4	P24G	0 V		
7	HI_COM1	Common		

As input signals, no-voltage contacts or transistor open collector inputs are used.

No-voltage contact specifications:
Contact rating: DC 24 V, 10 mA or over (circuit current: Approx. 7 mA)
Minimum contact current: DC 24 V, 1 mA
Contact impedance: 100 Ω or less

Transistor specifications:
Withstand voltage between collector and emitter: 30 V or over
Current between collector and emitter: 10 mA or over (circuit current: Approx. 7 mA)
Leak current between collector and emitter: 100 μA or less

b) Output signal Connector HAND

Pin	Signal name		Signal No.	Output circuit and example of connections
5	HO_1	Hand output 1	201	
12	HO_2	Hand output 2	202	
6	HO_3	Hand output 3	203	
13	HO_4	Hand output 4	204	
7	P24G	DC 24 V power supply GND		
14	P24G	DC 24 V power supply GND		

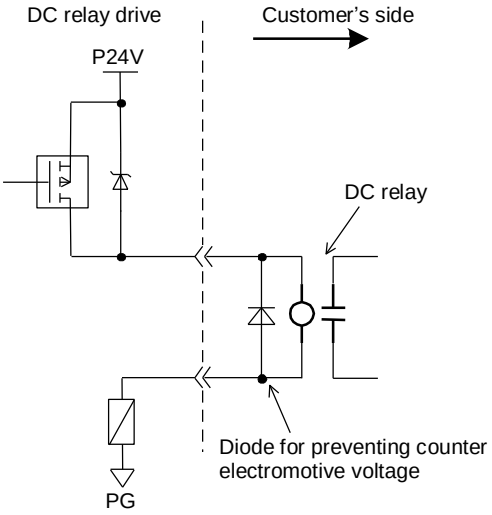
By using the DC 24 V power of the controller, a relay or solenoid valve, etc., can be driven. When the external power is used, GND of the external power should be common to GND (PG) of the TS1000 robot controller.

Output specifications:

Rated voltage : DC 24 V (Max. DC 30 V)
Rated current : 1 A (1.2 A or less including other I/Os when the internal power supply is used.)
Leak current : 100 µA or less

- If the DC 24 V power supplied from the robot controller is used, the total current should be 2 A or less.
- When a relay or solenoid valve, etc., is connected, it is necessary to use a surge killer or diode to absorb the surge voltage.
- When a double solenoid is used, HO_1 and HO_2, and HO_3 and HO_4 are used as pairs.

The following figure shows a DC relay circuit when the external power is used.



4.3 Permissible Load Conditions and Program Setting

This paragraph describes the permissible load conditions of the robot and program setting procedures according to the load.

4.3.1 Permissible Load Conditions

The robot load conditions are defined by the tool mass, moment of inertia and "tool center of gravity" offset value from the center of the tool shaft, as shown in Fig. 4.4. The load conditions allowed for TH250 and TH350 are given in Table 4.1 below.

Table 4.1 Permissible load conditions

Type of robot	Conditions	Permissible values
TH250/350	Mass of load	Max. 3 kg
	Moment of inertia	Max. 0.0170 kg·m^2
	"Load center of gravity" offset	Max. 70 mm

CAUTION

- NEVER use the robot under load conditions exceeding the permissible values. Otherwise, the robot life and safety cannot be guaranteed.

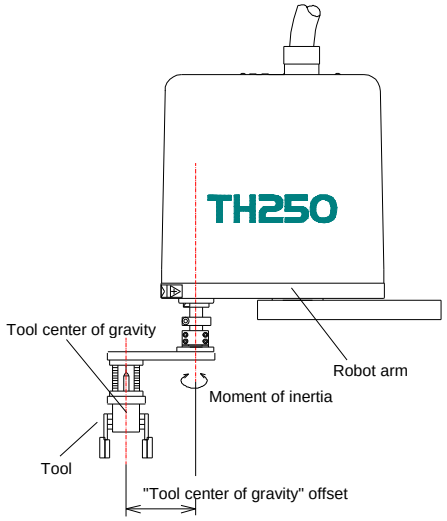


Fig. 4.4 Robot tool

4.3.2 Load Conditions and Program Setting

Even if the tool load conditions are within the permissible values listed in Table 4.1, appropriate acceleration/deceleration time and servo gain should be specified. If the setting of these values is inadequate, the robot may vibrate or overshoot, and will result in reduction in the robot life.
The TH series can automatically change the acceleration/deceleration time and servo gain by using the PAYLOAD command in the program according to the load conditions. The specific method for using this function is stated below.

a) PAYLOAD command
The programming format of the PAYLOAD command is as follows, when the tool mass is M kg and the "center of gravity" offset is L mm.

PAYLOAD = {M, L}
M : Mass of load (kg)
L : "Center of gravity" offset (mm)

The PAYLOAD command has the following functions.
- The acceleration/deceleration time of each robot axis is automatically changed according to the specified load conditions.
- The servo gain of each robot axis is automatically changed according to the specified load conditions.

b) Example programs
The basic example programs using the PAYLOAD command are shown below. For further information, see the Language Manual.

(Example program 1)
When moving the robot under load conditions of 2 kg load mass and 30 mm "center of gravity" offset:

```
PROGRAM SAMPLE
  SPEED= 100
  PAYLOAD= {2,30}
  MOVE A
  MOVE B
  STOP
END
```

(Example program 2)

Assume that when the robot grasps a workpiece while the hand mass is 1 kg and the "center of gravity" offset is 30 mm, the mass and "center of gravity" offset become 3 kg and 50 mm, respectively. When the robot picks and places workpieces under this condition:

```
PROGRAM SAMPLE
  PAYLOAD={1,30}
  ACCUR=COARSE
  ENABLE NOWAIT
  RESET DOUT
  MOVE TAIKI
  DOUT(1)
  WAIT DIN(1)
LOOP:
  MOVE A+POINT(0,0,100)
  IF DIN(-1)THEN GOTO FIN
  MOVE A
  WAIT MOTION>=100
  DOUT(213)
  DELAY 1
  PAYLOAD={3,50}
  MOVE A+POINT(0,0,100)
  MOVE B+POINT(0,0,100)
  MOVE B
  WAIT MOTION>=100
  DOUT(-213)
  DELAY 1
  PAYLOAD={1,30}
  MOVE B+POINT(0,0,100)
  GOTO LOOP
FIN:
  MOVE TAIKI
  DOUT(1)
  STOP
END
```

c) Setting of PAYLOAD command

In the initial state (i.e., unless the PAYLOAD command is used), the acceleration/deceleration and servo gain are set to 100 % and a value under minimum load, respectively. (See Para. 4.3.3.)

CAUTION

Be sure to use the PAYLOAD command in the following conditions.

- When the mass of load exceeds 1 kg.
- When there is a "load center of gravity" offset.

Unless the PAYLOAD command is used, the robot will vibrate or overshoot, and life of the robot mechanism will be shortened.

CAUTION

The moment of load inertia must be within the tolerance given in Table 4.1. If the moment of inertia is large though there is no "load center of gravity" offset, the robot may vibrate. When this happens, figure out the virtual "center of gravity" offset (L mm) from the following equation, using the moment of inertia (J kg·m^2) and mass (M kg).

$$L = \sqrt{J \times 10^6 / M}$$

Then, designate the following.

PAYLOAD={M, L}

CAUTION

When guiding the robot manually, if the mass of load or the "center of load gravity" offset is large, the robot may vibrate due to inappropriate servo gain. When this happens, execute the following operation while selecting the function of load conditions in the test run mode.

DO PAYLOAD={5,100}

EXE

Then, the servo gain satisfying the load conditions is selected.

If the robot still vibrates or overshoots after setting the above conditions, consult with SHIBAURA MACHINE. Also clarify the load conditions and program then.

4.3.3 Setting of Acceleration/Deceleration Time for Load Conditions

a) TH250

When the PAYLOAD command is used, the acceleration/deceleration time of the robot is automatically changed according to the load conditions.
For the mass of load, the acceleration/deceleration time changes as shown in Fig. 4.5. The vertical line shows the acceleration during acceleration/deceleration. If the mass of load is 3 kg, for instance, the acceleration is set to 73 % of the maximum.

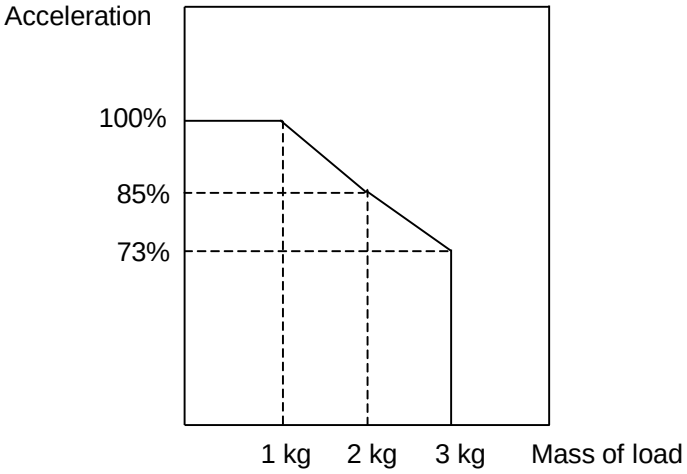


Fig. 4.5 Setting of acceleration to mass of load (TH250)

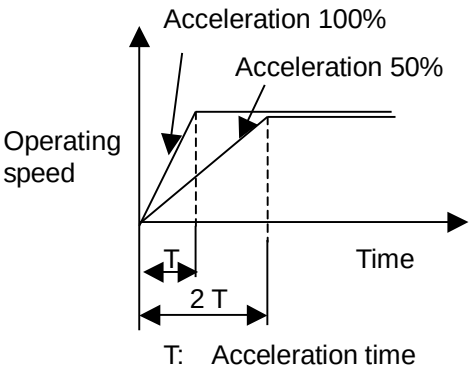


Fig. 4.6 Change of acceleration

Additionally, if there is a "center of load gravity" offset, the acceleration/deceleration time changes as shown in Fig. 4.7.
If the mass of load is 3 kg and the "center of gravity" offset is 70 mm, the acceleration is set to 25 % (= 73 % × 34 %) of the maximum.

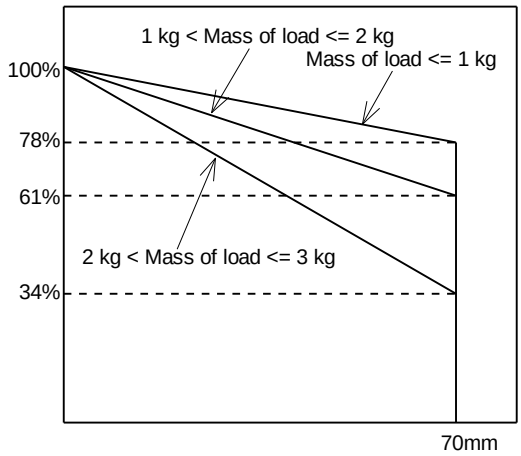


Fig. 4.7 Setting of acceleration for "center of gravity" offset (TH250)

b) TH350

When the PAYLOAD command is used, the acceleration/deceleration time of the robot is automatically changed according to the load conditions.
For the mass of load, the acceleration/deceleration time changes as shown in Fig. 4.8. The vertical line shows the acceleration during acceleration/deceleration. If the mass of load is 3 kg, for instance, the acceleration is set to 70 % of the maximum.

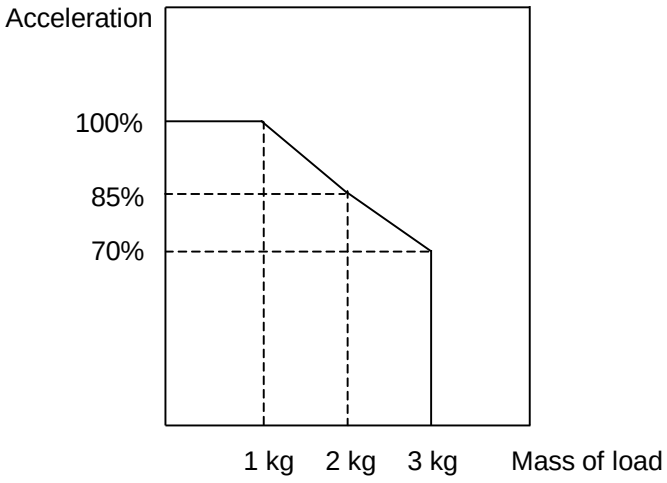


Fig. 4.8 Setting of acceleration for mass of load (TH350)

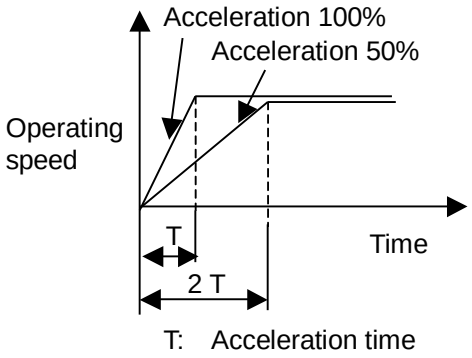


Fig. 4.9 Change of acceleration

Additionally, if there is a "center of load gravity" offset, the acceleration/deceleration time changes as shown in Fig. 4.10.
If the mass of load is 3 kg and the "center of gravity" offset is 70 mm, the acceleration is set to 21 % (= 70 % × 30 %) of the maximum.

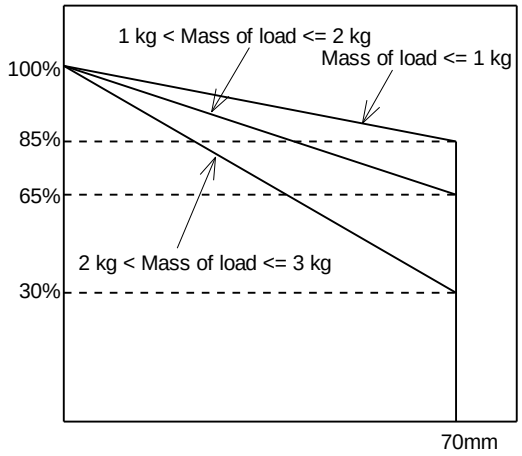


Fig. 4.10 Setting of acceleration for "center of gravity" offset (TH350)

Section 5

Specifications

5.1 Specifications Table

Item		Specifications	
Structure		Horizontal multi-joint type SCARA robot	
Model		TH250	TH350
Applicable controller		TS1000	
Mass of actuator		15 kg	16kg
No. of controlled axes		Four (4)	
Arm length		250 mm (125 mm + 125 mm)	350 mm (225 mm + 125 mm)
Motor capacity	Axis 1	200 (W)	
	Axis 2	100 (W)	
	Axis 3	100 (W)	
	Axis 4	100 (W)	
Operating range	Axis 1	±115 (deg)	
	Axis 2	±140 (deg)	
	Axis 3	120 (mm)	
	Axis 4	±360 (deg)	
Maximum speed (*1)	Axis 1	480 (deg/s)	300 (deg/s)
	Axis 2	480 (deg/s)	
	Axis 3	1067 (mm/s)	
	Axis 4	1143 (deg/s)	
	Composite speed of axes 1 and 2	3.14 (m/s)	2.88 (m/s)
Rated payload mass		1 (kg)	
Maximum payload mass		3 (kg)	
Permissible load inertia (*1)		0.017 (kg·m^2)	
Repeatability	X, Y	±0.01 (mm)	
	Z	±0.01 (mm)	
	C	±0.03 (deg)	
Cycle time (*2) (When payload mass is 2 kg)		0.49 (sec)	

Drive system	By means of AC servo motors
Position detection method	Absolute

*1: When the mass of load exceeds 1 kg, or when the gravity center position of load is away from the axis 4 center position, both the speed and acceleration should be reduced, using the PAYLOAD command.

*2: Shuttle time for rough positioning in horizontal direction of 300 mm and vertical direction of 25 mm.

! CAUTION

Put the Z-axis (axis 3) in the raised position as much as possible, when moving Axes 1, 2, and 4.

Moving Axis 1, 2, or 4 when the Z-axis is in low positions can lead to premature damage to the ball screw spline (Z-axis shaft).
If Axis 1, 2, or 4 must be moved while the Z-axis is in low positions due to unavoidable circumstances, prevent the ball screw spline vibrations by using the SPEED, ACCEL/DECEL, and PAYLOAD commands to adjust the operation speed and acceleration.
If Axis 1, 2, or 4 is moved while the Z-axis is in low positions, be extremely careful not to collide with any objects. Even if Axis 1, 2, or 4 is moved at low speed, a collision or other impact with an object can damage the ball screw spline (Z-axis shaft) before alarm occurs.

Appendix A Calculating Control Power Capacity

The power capacity of the controller can be calculated in the following manner. The power connected to the controller should exceed the capacity obtained from the following equation:

(Power capacity) = (Power capacity of control section) + (Power capacity of servo driver)

a) Power capacity of control section
The power capacity of the controller excluding the servo driver is as follows:
Power capacity of control section ..... 306 VA

b) Power capacity of servo driver
The power capacity of the servo driver is the total of the power capacities of the servomotors used in the robot. The following table lists the power capacity of each motor.

Motor capacity (W)	Power capacity (VA)
100	179
200	356

Example: The servo motor capacity and servo driver power capacity of the TH series robots are as follows:

Type of robot	Axis 1	Axis 2	Axis 3	Axis 4	Power capacity of servo driver (VA)
TH250	200	100	100	100	356 + 179 + 179 + 179 = 893
TH350	200	100	100	100	356 + 179 + 179 + 179 = 893

Reference: The power capacity of each servo motor is calculated by the following equation.

(Power capacity of servo driver) = $\sqrt{3} * (E * I)/(\cos\theta * \eta)$
where,
E : Motor rated voltage
I : Motor rated current
Cos θ : Servo driver power factor (0.7)

η : Servo driver power factor (0.85)

c) Example of calculating power capacity

The power capacity of TH series robot is listed in the following table.

Type of robot	Power capacity of control section (VA)	Power capacity of servo driver (VA)	Power capacity of controller (VA)
TH250	306	893	1199
TH350	306	893	1199

Note: With respect to the power capacity required, it is assumed that all axes are continuously operated at 100% load. If the power capacity is insufficient when the robot is operated, troubles such as a drop in power voltage, a malfunction, or resetting of the control power may take place. Thus, the robot may not be correctly operated.

Also, a current three (3) times the rated current may be required at the start of the robot. Thus, it is necessary to select a power supply in which the peak current of the power is three (3) times the rated current for 100 ms.